

Operating System [Unit 4]

Lab 4

Deadline: 2081-05-10

1. Implementation of First fit Memory Allocation Strategies.
2. Implementation of Next fit Memory Allocation Strategies.
3. Implementation of Best fit Memory Allocation Strategies.
4. Implementation of Worst fit Memory Allocation Strategies.
5. Implementation of FIFO Page Replacement Algorithm.
6. Implementation of OPR Page Replacement Algorithm.
7. Implementation of LFU Page Replacement Algorithm.
8. Implementation of LRU Page Replacement Algorithm.
9. Implementation of Second-Chance Page Replacement Algorithm.

Assignment 4

Deadline: 2081-05-15

1. Explain memory management with example.
2. What is page fault and page hit? Discuss about Memory Allocation Strategies.
3. Explain with example about Belady's Anomaly.
4. How Second Chance Algorithm differs from FIFO policy.
5. Why is OPR best but not practically feasible?
6. Differentiate:
 - a. Segmentation and Fragmentation
 - b. Internal and External Fragmentation
 - c. Paging and Segmentation
 - d. Page table and Segmentation table
7. What is virtual memory? Discuss how large files can execute on low memory.
8. Discuss in detail the use of TLB in the paging.
9. How Hash table works in paging.
10. Explain Segmentation with MULTICS.
11. Explain Associative memory and Virtual Machines.

12. Explain free space management.

13. What is primary memory management? Explain mapping of virtual address with real address.